

NYC 311

OPERATIONAL STRAIN **ANALYSIS**

DATA ANALYSIS WITH PYTHON AND SQL | FINAL PROJECT

When does 311 demand become operational strain?

COMPLAINT TYPE

GEOGRAPHY

SEASONALITY

WEATHER

SEPARATING CITYWIDE TRENDS FROM INDIVIDUAL COMPLAINT BEHAVIOR

Data Architecture

Full 2025 Complaint Data Set

True 2025 daily complaint counts for total complaint volume

Complaint-Level Analysis Sample

~800 complaints/month for complaint-level detail

NOAA Weather

Daily weather data merged via SQL

MERGED SQL DATASET

- Volume analyses → *Full 2025 Complaint Data Set*
- Complaint characteristics & resolution time → *Complaint-Level Analysis Sample*

The FULL dataset was used whenever measuring total complaint volume. Meanwhile, the sample was used to analyze complaint-level resolution behavior (stratified daily sampling to avoid start-of-month bias)

COMPLAINT LANDSCAPE

COMPLAINT MIX VARIES BY BOROUGH

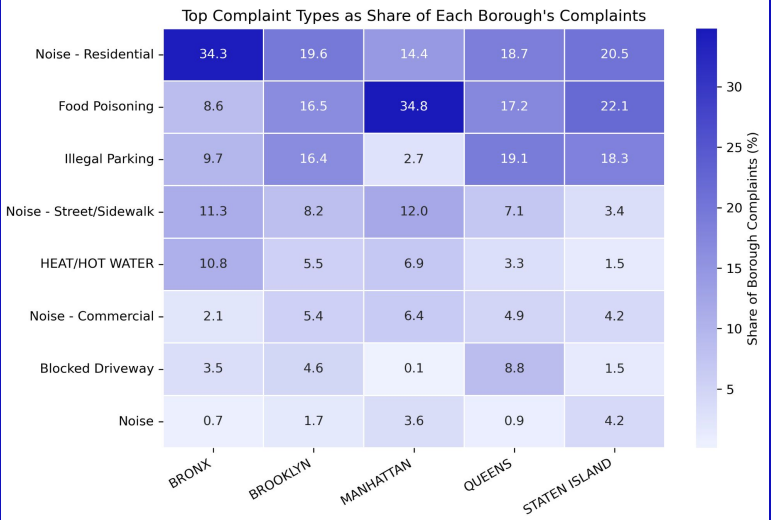
BUT THE SAME CORE CATEGORIES APPEAR CITYWIDE

Top Complaint Types by Sampled Volume | Complaint-Level Sample Dataset

Complaint Type	Sampled Volume	Median Hrs to Close	Speed	Cross-Borough Variation
Noise – Residential	2,030	1.2 hrs	●	1.2 hrs
Food Poisoning	1,895	72.0 hrs	●	185.7 hrs
Illegal Parking	1,171	1.9 hrs	●	1.9 hrs
Heat/Hot Water	609	39.0 hrs	●	17.8 hrs
Unsanitary Condition	120	240.9 hrs	●	200.8 hrs

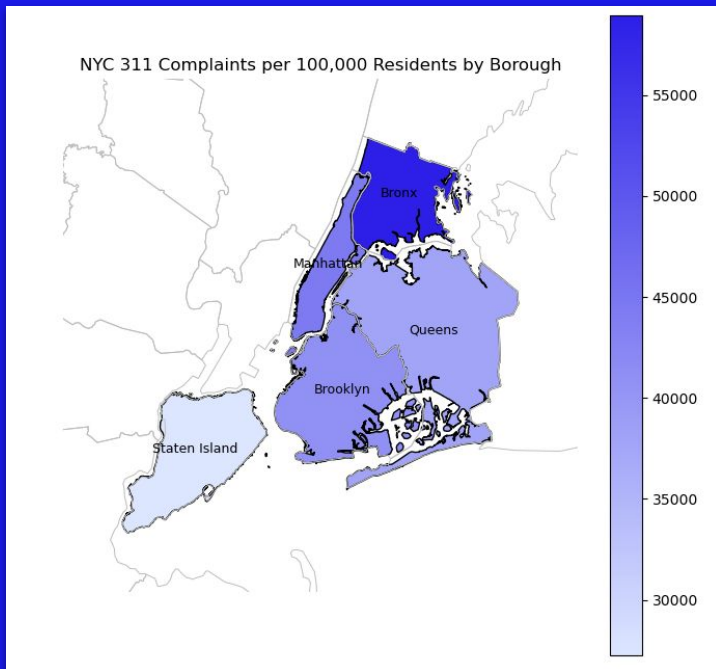
● Fast (<5 hrs) ● Medium (5–50 hrs) ● Slow (>50 hrs)

VOLUME SHARE % BY BOROUGH



BOROUGH ANALYSIS

BROOKLYN HAS THE MOST COMPLAINTS, BUT THE BRONX HAS THE HIGHEST COMPLAINT BURDEN PER RESIDENT



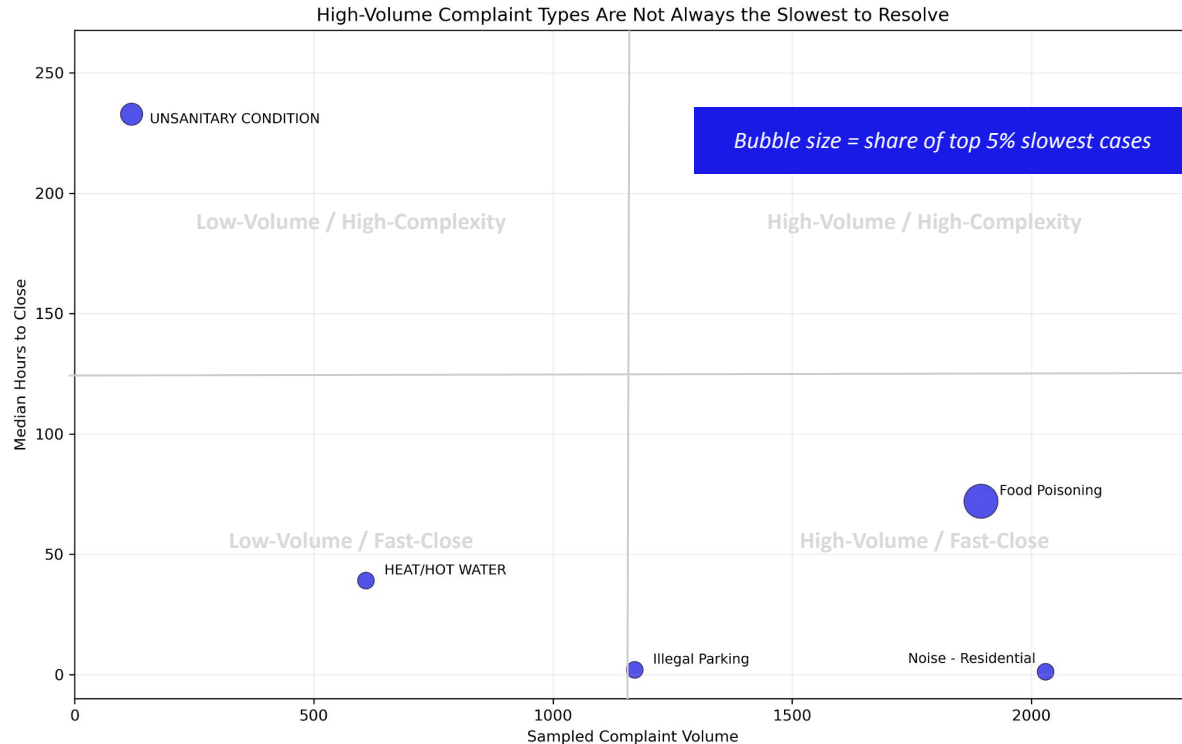
Borough	True Complaints	Population	Per 100k Residents
<u>Bronx</u>	836,505	1,419,250	58,940
Manhattan	725,036	1,627,788	44,541
<u>Brooklyn</u>	1,078,327	2,646,306	40,748
Queens	878,212	2,330,124	37,689
Staten Island	134,302	492,734	27,256

Population size explains 92.6% of the variation in raw complaint counts across boroughs (Pearson $r = 0.926$, Exec 67). Volume almost perfectly tracks how many people live somewhere, not how strained a borough's services are. Normalizing by population exposes the Bronx as the true operational priority.

Equitable service delivery demands prioritizing the Bronx, not simply the most populous boroughs

Resolution time varies by 125× across complaint types

Complaint Types Driving Extreme Delays



OPERATIONAL IMPLICATION

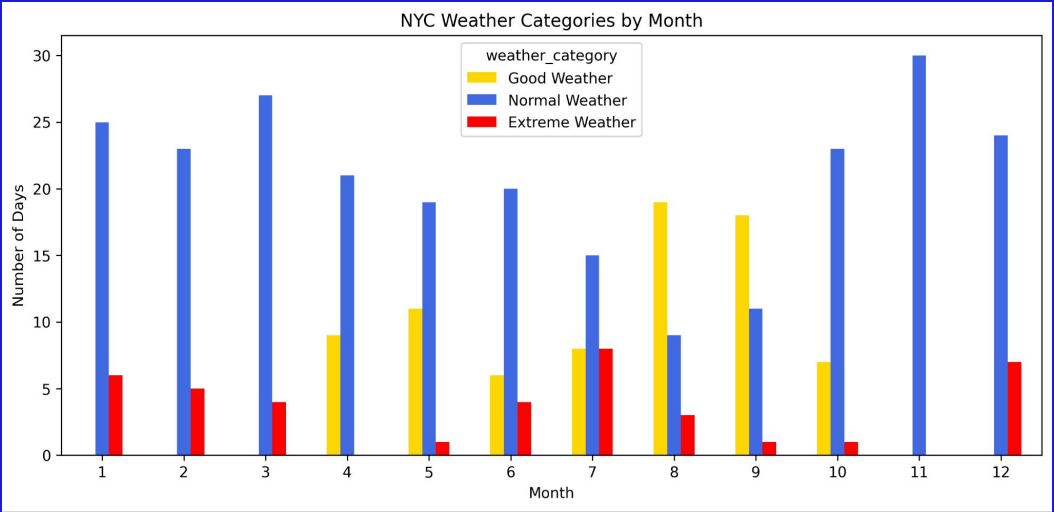
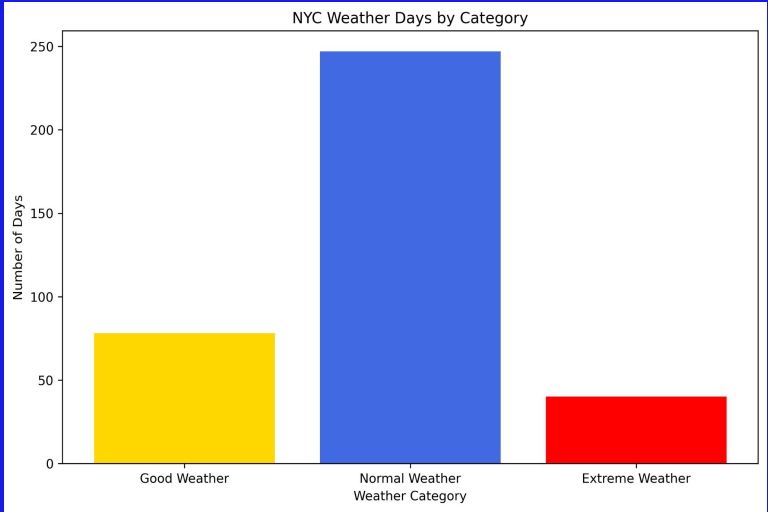
Unsanitary Condition complaints require physical inspections, specialized crews, and often multi-agency coordination.

Resolution speed is driven by case complexity — not overall workload. Staffing interventions should target complaint type, not raw volume.

*Illegal Parking closes in ~2 hrs;
Unsanitary Condition takes ~239 hrs.*

WEATHER ANALYSIS

WEATHER AFFECTS RESOLUTION TIMES
MORE THAN TOTAL COMPLAINT VOLUME



Weather Category	Avg Daily Complaints	Median Hours to Close
Good Weather	~10,145	~6.6 hrs
Normal Weather	~9,851	~8.0 hrs
Extreme Weather	~10,694	~11.3 hrs

EXTREME WEATHER CHANGES COMPLAINT MIX MORE THAN TOTAL VOLUME

SQL Analysis: Complaint Types That Increase Disproportionately During Extreme Weather

Complaint Type	Normal Days (avg/day)	Extreme Weather Days (avg/day)	% Increase	Operational Note
HEAT/HOT WATER	3.0	4.2	+40%	Driven by extreme heat days; often requires housing inspections and coordination with building management
Food Poisoning	5.3	5.8	+9.4%	Extreme heat accelerates food spoilage
Illegal Parking	3.4	3.3	-2.9%	Remains relatively stable; likely given incidence of driving does not change with weather
Noise – Residential	5.7	4.7	-15.8%	Fewer people outdoors during extreme events

HEAT DAYS → SURGE PROTOCOL

Pre-position sanitation and housing inspection crews on extreme heat days — not a general staffing surge.

TARGETED, NOT CITYWIDE

Extreme weather doesn't increase total complaints — it shifts the mix toward structurally slower-to-close types.

COMPLAINT DAY CLASSIFICATION

**BAD DAYS ARE NOT RANDOM, THEY PERSIST —
AND THEY'RE NOT JUST WEATHER EVENTS**

	Bad Complaint Day	Normal Complaint Day	Good Complaint Day
Good Weather	0	77	1
Normal Weather	20	213	14
Extreme Weather	11	26	3

Cross-Tab: Weather Category vs Complaint-Day Type (Exec 155)

Complaint Day Type	Avg Volume	Median Hrs to Close
Bad Complaint Day	12,344	20.15 hrs
Normal Day	9,854	7.3 hrs
Good Complaint Day	8,660	1.1 hrs

BAD DAYS PERSIST

After a bad day, the probability of another bad day is ~6.5× the baseline. The system typically self-corrects within one day — but that window matters for staffing decisions.

**~6.5× recurrence risk
after a bad day**

Complaint mix, not weather, defines bad days. Many of the worst operational days occur under normal or good weather.

Complexity drives delays, not volume

WORST DAYS ANALYSIS

THE WORST OPERATIONAL DAYS OF 2025
MOSTLY HAPPENED IN GOOD OR NORMAL WEATHER

Top 5 Slowest-Resolution Days of 2025 (Exec 163)

Date	Median Hrs to Close	Weather Category	Daily Volume
Aug 18, 2025	234.6 hrs	Good Weather ★	10,152
Aug 15, 2025	84.3 hrs	Normal Weather	9,930
Aug 21, 2025	71.6 hrs	Normal Weather	9,339
May 8, 2025	68.5 hrs	Good Weather ★	9,957
Dec 27, 2025	60.3 hrs	Extreme Weather	7,833

★ Good Weather rows highlighted | Ordered by median hours to close

THE REBUTTAL

The single worst resolution day of 2025 — August 18 — had a median close time of 234.6 hours.

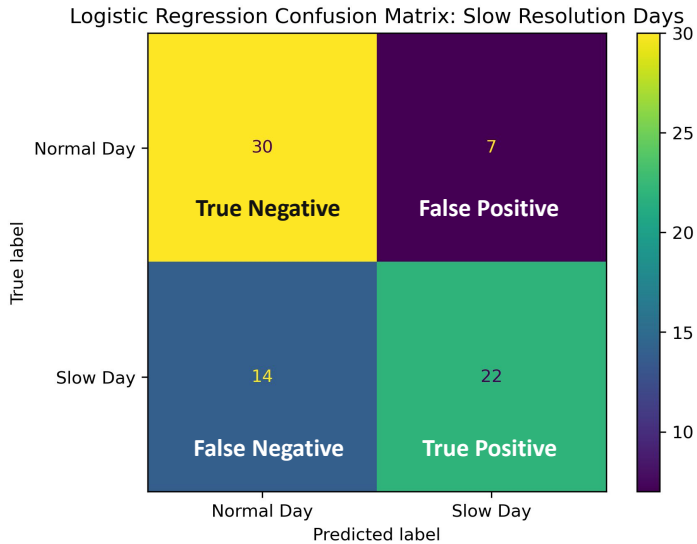
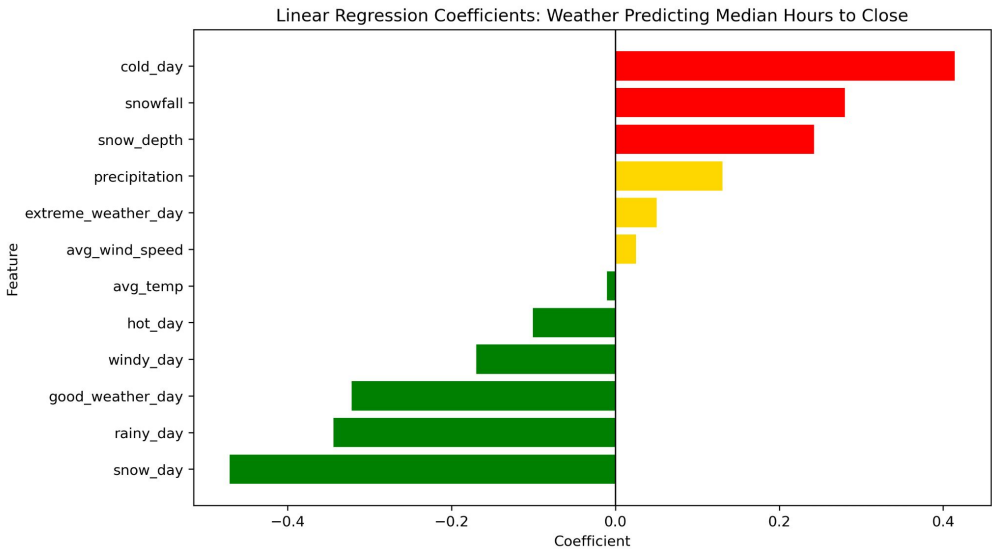
It was a Good Weather day.

64.5% of all Bad Complaint Days occur under Normal weather. Complaint mix — not weather — is the operational driver.

Weather-based alerts alone miss most operational failures

WEATHER PREDICTS SLOW DAYS — BUT ITS ONLY PART OF THE STORY

Linear + Logistic Regression



Model	Metric	Value	Interpretation
Linear Reg.	R ²	0.316	Weather explains ~31.6% of resolution time variance
Logistic Reg.	Accuracy	71.2%	Correctly classifies 52 of 73 test days
Logistic Reg.	Precision	75.9%	When it predicts a slow day, it is usually right
Logistic Reg.	False Neg.	14 / 36	Missed 14 actual slow days, suggesting non-weather drivers still matter

NYC 311 STRAIN IS A *COMPLEXITY PROBLEM,* NOT A VOLUME PROBLEM.

01

TARGET COMPLAINT TYPE, NOT VOLUME

Unsanitary condition and building/use complaints drive the most extreme delays as they require inspections, specialist crews, and multi-agency coordination that no staffing surge can shortcut. Fix the workflow, not the headcount.

02

THE BRONX IS THE EQUITY PRIORITY

The Bronx carries the highest complaints per 100k residents of any borough. Allocating resources by raw complaint counts systematically underserves Bronx residents. Per-capita burden must drive the budget.

03

WEATHER IS A SIGNAL, NOT THE STORY

Weather explains ~32% of resolution-time variance and predicts slow days with ~71% accuracy. The remaining 68% is complaint mix, staffing, and agency capacity. Use weather as a one-day-ahead trigger, not a standalone explanation.